

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study
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GitHub Repository	https://github.com/Amsavarthini25/ML_LAB_2470012

1 Objective

- To implement a Decision Tree classifier.
- To extend the Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare single-tree and ensemble-tree models.

2 Problem Statement

Given the Wisconsin Diagnostic Breast Cancer dataset, the task is to classify each tumor sample as Malignant or Benign using the 30 numerical features describing cell nuclei characteristics. A single Decision Tree classifier and a Random Forest ensemble are trained on an 80–20 train-test split, tuned using 5-Fold Cross-Validation over their respective hyperparameter spaces, and compared using standard classification metrics to study how ensembling affects overfitting and generalization.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer Dataset
Dataset Source	<code>sklearn.datasets.load_breast_cancer</code> (UCI Machine Learning Repository)
Number of Samples	569
Number of Features	30 numerical attributes
Number of Classes	2 (Malignant, Benign)
Missing Values	None
Train-Test Split	80–20 (455 training samples, 114 testing samples), stratified

4 Exploratory Data Analysis

Class Distribution

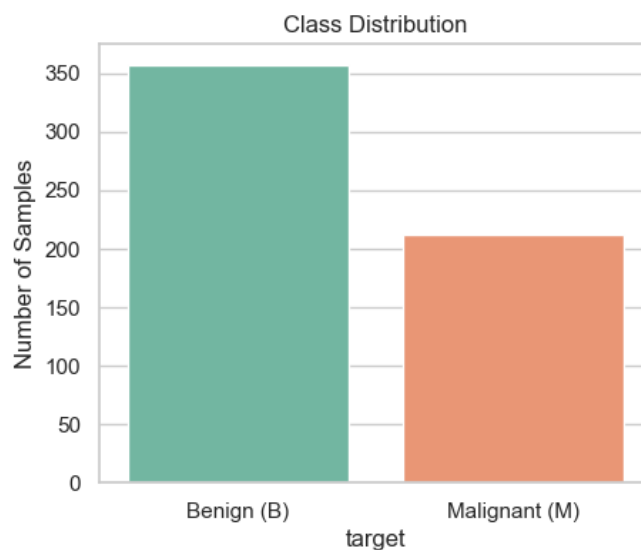


Figure 1: Class distribution of Malignant and Benign samples

Inference:

The dataset has 357 benign cases against 212 malignant ones, so it's mildly imbalanced but not to a worrying degree. This is why I went with a stratified train-test split, so both classes stay proportionally represented in training and testing. It also means accuracy alone isn't fully trustworthy, so precision, recall and F1 matter more here.

Feature Correlation Analysis

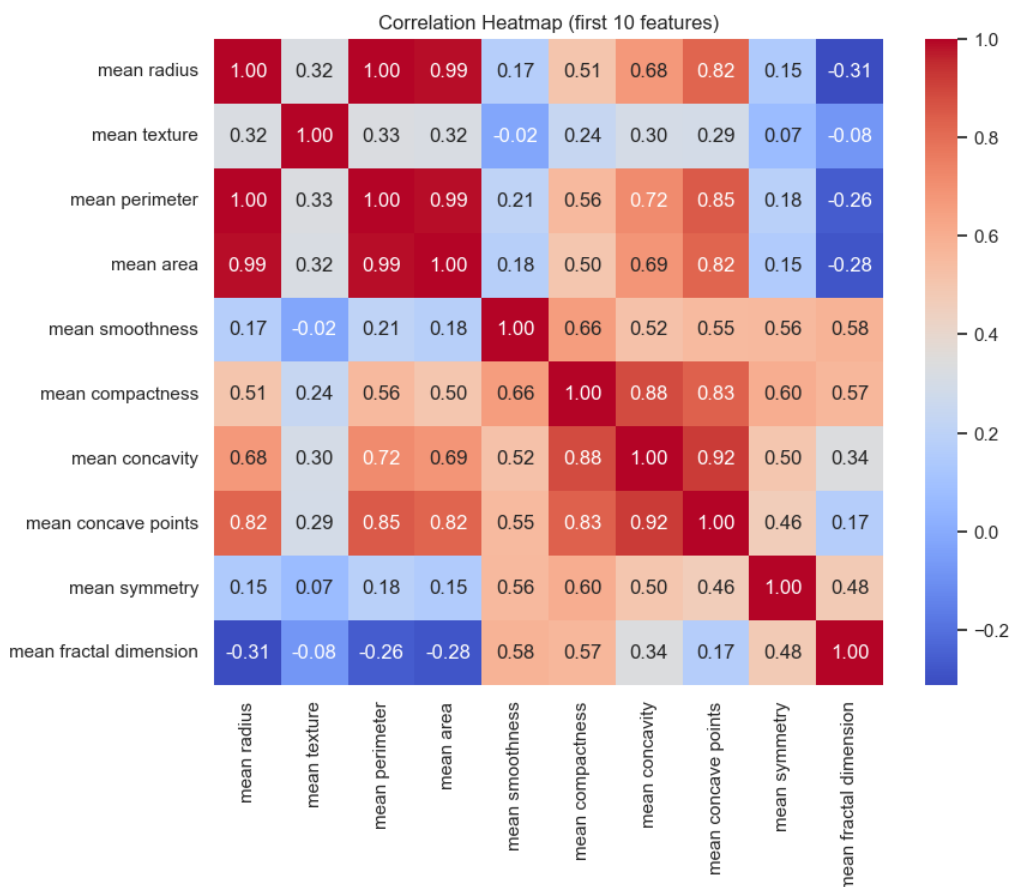


Figure 2: Correlation heatmap of the first 10 features

Inference:

A lot of the size-related features like radius, perimeter and area are very strongly correlated with each other, which makes sense since they're all measuring the same physical property in different ways. Features like smoothness and symmetry show weaker correlation with the size-based ones, so they likely carry more independent information. This kind of redundancy is something tree-based models handle naturally since they just pick the most useful feature to split on at each node.

5 Data Preprocessing

- **Missing value handling:** The dataset has no missing values, so no imputation was required.
- **Encoding:** Class labels are already numerically encoded by scikit-learn (0 = malignant, 1 = benign).
- **Feature scaling:** Not required, since Decision Trees and Random Forests are scale-invariant (splits are based on feature thresholds, not distances).
- **Train-test split:** 80–20 stratified split using `train_test_split` with `random_state=42`.

- **Feature engineering:** None applied; all 30 original numerical features were used directly.

6 Machine Learning Algorithm

Decision Tree Classifier

- **Working principle:** A Decision Tree recursively partitions the feature space by selecting, at each node, the feature and threshold that best separates the classes according to an impurity measure (Gini Index or Entropy). This continues until a stopping criterion (max depth, minimum samples per split/leaf) is reached.
- **Mathematical formulation:** Gini Index: $Gini = 1 - \sum_{i=1}^c p_i^2$; Entropy: $H = - \sum_{i=1}^c p_i \log_2(p_i)$, where p_i is the proportion of class i samples at a node.
- **Advantages:** Easy to interpret, requires no feature scaling, handles non-linear relationships naturally.
- **Limitations:** Prone to high variance and overfitting, especially when grown to full depth.

Random Forest Classifier

- **Working principle:** Random Forest builds an ensemble of Decision Trees, each trained on a bootstrapped sample of the training data and considering only a random subset of features at each split. Final predictions are made by majority voting across all trees.
- **Mathematical formulation:** $\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_T(x)\}$, where $h_t(x)$ is the prediction of the t -th tree.
- **Advantages:** Reduces variance and overfitting compared to a single tree, generally more robust and accurate.
- **Limitations:** Less interpretable than a single tree, more computationally expensive, larger memory footprint.

7 Experimental Setup

Python Version	3.x
Scikit-Learn Version	As per environment
IDE	Jupyter Notebook
Operating System	Windows
Hardware	Standard CPU

8 Hyperparameter Selection

Decision Tree Hyperparameter Search Space

Explored: `criterion` $\in$ {gini, entropy}, `max_depth` $\in$ {3, 5, 7, None}.

Table 2: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
gini	3	92.53	0.9400
gini	5	92.31	0.9372
gini	7	92.31	0.9375
gini	None	91.65	0.9319
entropy	3	93.41	0.9480
entropy	5	92.97	0.9439
entropy	7	92.97	0.9440
entropy	None	92.97	0.9440

Best Decision Tree hyperparameters: `criterion` = entropy, `max_depth` = 3
(Avg CV Accuracy = 93.41%, Avg CV F1 = 0.948).

Random Forest Hyperparameter Search Space

Explored: `n_estimators` $\in$ {50, 100}, `max_depth` $\in$ {5, 10}, `max_features` $\in$ {sqrt, log2}.

Table 3: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
50	5	sqrt	95.60	0.9647
50	5	log2	95.60	0.9651
50	10	sqrt	95.60	0.9648
50	10	log2	96.04	0.9685
100	5	sqrt	96.04	0.9685
100	5	log2	95.82	0.9668
100	10	sqrt	96.26	0.9699
100	10	log2	96.48	0.9718

Best Random Forest hyperparameters: `n_estimators` = 100, `max_depth` = 10,
`max_features` = log2 (Avg CV Accuracy = 96.48%, Avg CV F1 = 0.9718).

9 5-Fold Cross-Validation Performance Comparison

Table 4: 5-Fold Cross-Validation Accuracy Comparison (%)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	90.11	94.51	91.21	94.51	96.70	93.41
Random Forest	96.70	97.80	93.41	95.60	98.90	96.48

10 Results

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	0.9474	0.9459	0.9722	0.9589
Random Forest	0.9561	0.9589	0.9722	0.9655

11 Result Visualization

Confusion Matrices

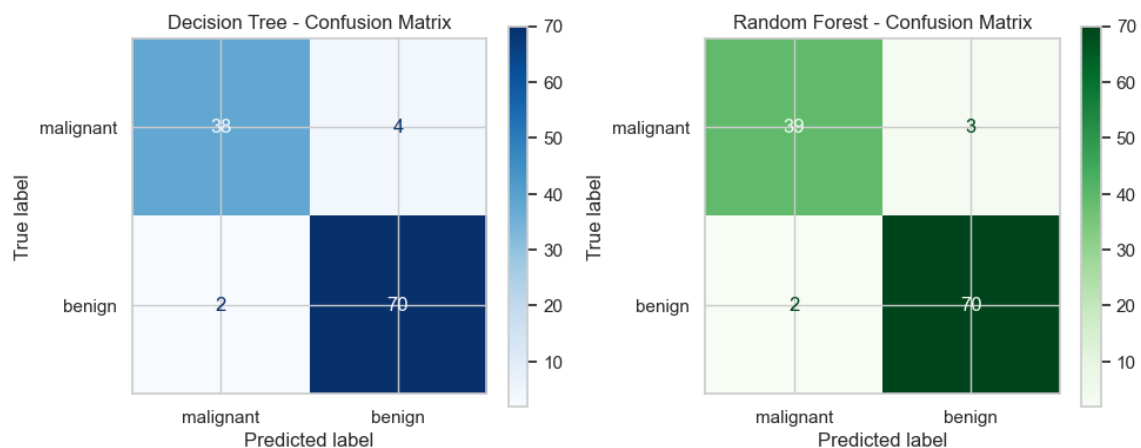


Figure 3: Confusion matrices for Decision Tree and Random Forest on the test set

Inference:

Both models correctly identify most benign cases, but the Random Forest makes fewer mistakes overall on the malignant class. The false positives are slightly reduced when moving from the single tree to the ensemble. Since this is a medical diagnosis problem, even small reductions in misclassification matter, especially for the malignant class where missing a case is costlier.

ROC Curve and AUC

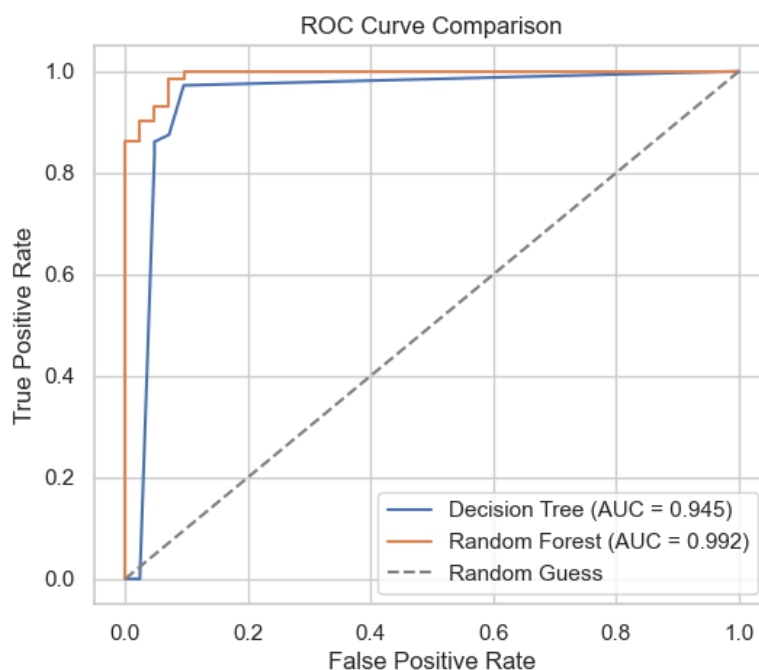


Figure 4: ROC curves for Decision Tree and Random Forest

Inference:

The Random Forest's ROC curve sits closer to the top-left corner and has a higher AUC than the Decision Tree, meaning it separates the two classes better across all thresholds. The single tree's curve is a bit more jagged, which is typical since it only has a few possible decision boundaries. Overall, this confirms that averaging multiple trees gives smoother and more reliable probability estimates.

12 Conclusion

Working through this experiment showed me how a single Decision Tree can overfit once it's allowed to grow too deep, and how tuning something as simple as depth and criterion through cross-validation makes a real difference. Building the Random Forest on top of that made the improvement in stability and accuracy pretty obvious to see across the folds and metrics. It also helped me understand practically why bagging and random feature selection are supposed to reduce variance, not just as a theory but as something visible in the numbers. Overall this made the overfitting-vs-generalization trade-off feel a lot more concrete than just reading about it.